

Module 2

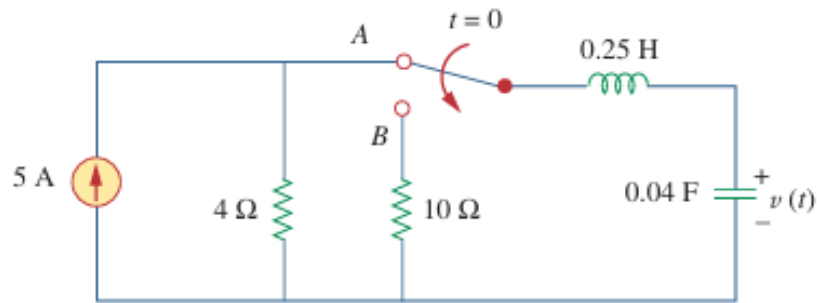
Transient Analysis of First and Second-Order Circuits

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Tutorial

1. If $R = 50 \, \Omega$, $L = 1.5 \, \text{H}$, what value of C will make an RLC series circuit: (a) overdamped, (b) critically damped, (c) underdamped?
2. In the circuit, the switch instantaneously moves from position A to B at $t = 0$. Find $v(t)$ for all $t \geq 0$.



3. The responses of a series RLC circuit are

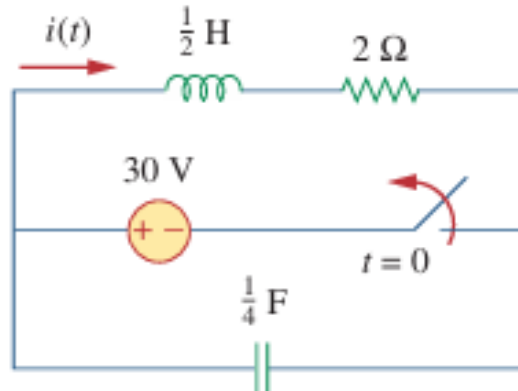
$$v_C(t) = 30 - 10e^{-20t} + 30e^{-10t} \, \text{V}$$

$$i_L(t) = 40e^{-20t} - 60e^{-10t} \, \text{mA}$$

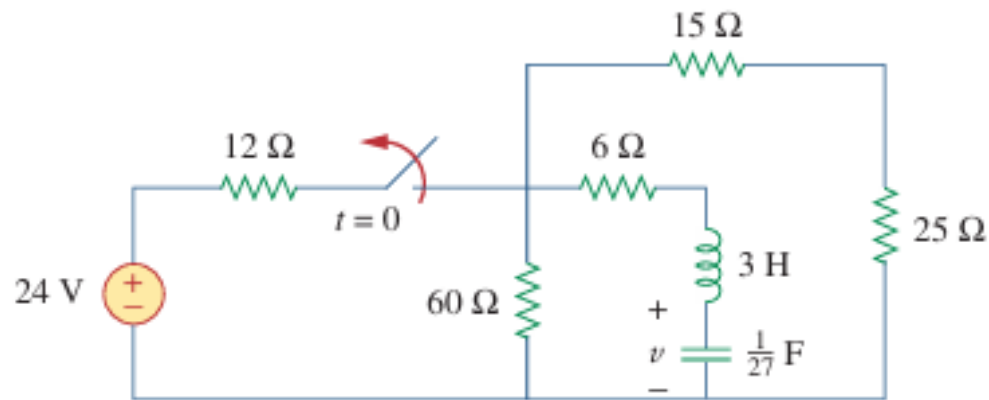
where v_c and i_L are the capacitor voltage and inductor current, respectively. Determine the values of R , L , and C .

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4. The switch in the circuit has been closed for a long time but is opened at $t = 0$. Determine for $i(t)$ for $t > 0$.

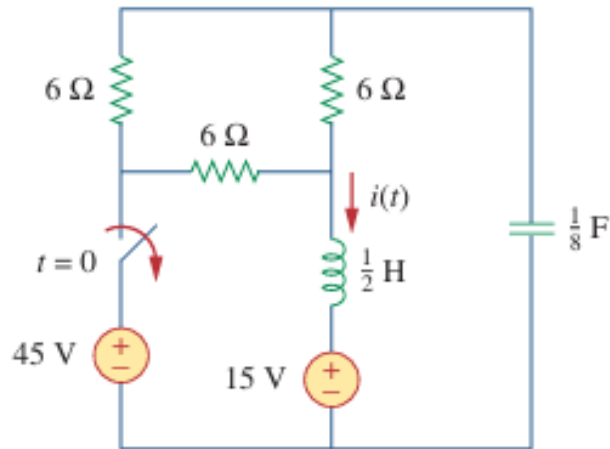


5. Calculate $v(t)$ for $t > 0$ in the circuit.

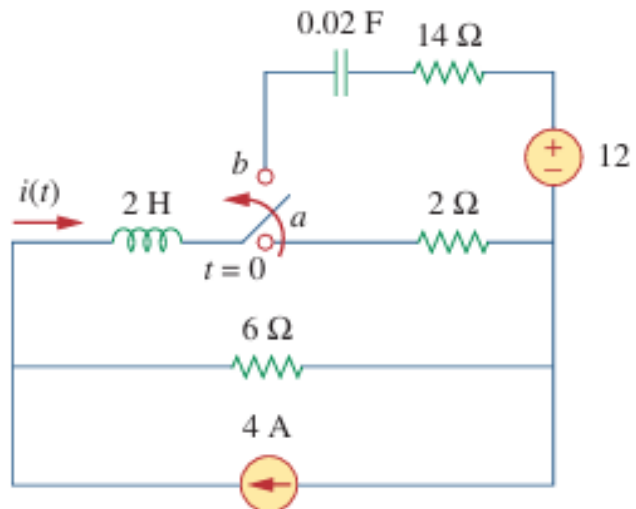


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6. For the network shown below, solve for $i(t)$ for $t > 0$.

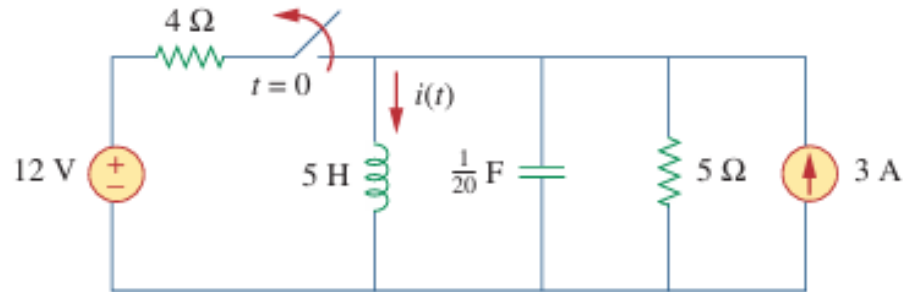


7. The switch in the circuit is moved from position a to b at $t = 0$. Determine $i(t)$ for $t > 0$.



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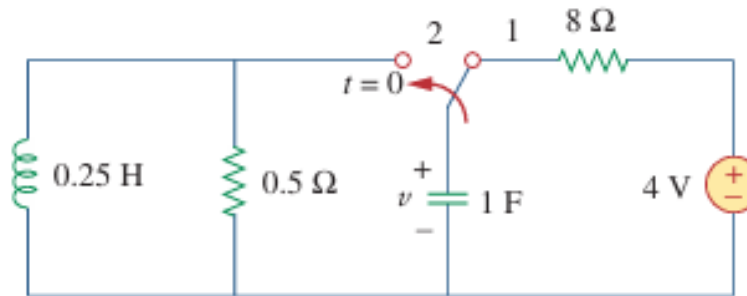
8. Determine $i(t)$ for $t > 0$ in the circuit.



9. In the circuit, the switch has been in position 1 for a long time but moved to position 2 at $t = 0$. Find:

(a) $v(0^+)$, $dv(0^+)/dt$,

(b) $v(t)$ for $t \geq 0$.



10. Find $v(t)$ for $t > 0$ in the circuit.

